

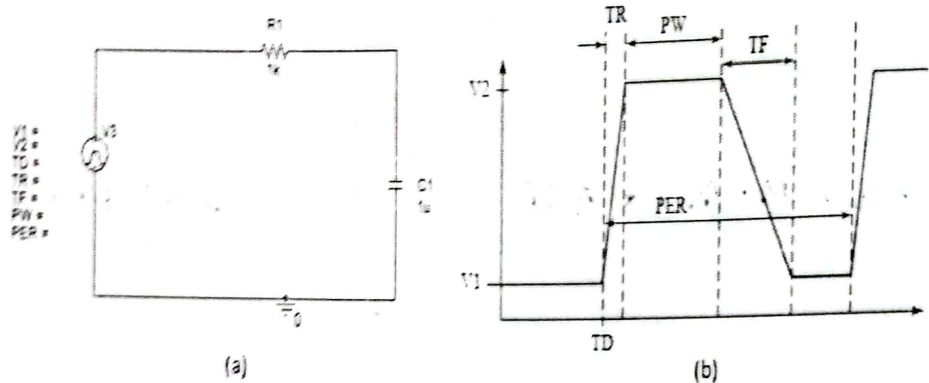


Figure 4.1: (a) Schematic of a simple RC circuit connected to pulse voltage waveform: (b) Diagram of the SPICE VPULSE parameter definitions

For the purposes of this discussion, we set a zero-time delay, $V_1 = 0$, and $V_2 = 9\text{ V}$. The circuit time constant is $\tau = RC = 1\text{ ms}$, so we set the rise and fall times to be 1 ns . Although SPICE will not allow a voltage to change in zero time since it solves the differential equations using discrete time intervals, compared to our circuit time constant 1 ns is a reasonable approximation to "instantaneous." We will consider four basic cases, summarized in Table 4.1. In the first two cases, the input-pulse width (PW) is much longer than the circuit time constant τ , so we expect the transients resulting from the beginning of the pulse to die out before the pulse is over. In the latter two cases, the opposite is true: the pulse width is so short that the capacitor does not have time to fully charge before the pulse ends. A similar issue arises when we consider the response of the circuit when the time between pulses ($PER - PW$) is either short (Case II) or long (Case III) compared to the circuit time constant.

Table 4.1 - Four separate cases of pulse width and period relative to the circuit time constant of 1 ms

Case	Pulse Width (PW)	Period (PER)	($PER - PW$) (2 Points)
I	10 ms ($\tau \ll PW$)	20 ms ($\tau \ll PER - PW$)	10 ms
II	10 ms ($\tau \ll PW$)	10.1 ms ($\tau \gg PER - PW$)	0.1 ms
III	0.1 ms ($\tau \gg PW$)	10.1 ms ($\tau \ll PER - PW$)	10 ms
IV	0.1 ms ($\tau \gg PW$)	0.2 ms ($\tau \gg PER - PW$)	0.1 ms

Task 2: To examine the response of RC Circuit

$$\tau_{\text{(Calculated)}} = 1000 \times \frac{\text{Pre-Lab (2 Points)}}{(1/1000000)} = 1\text{ ms}$$

Case I: When the capacitor has enough time to charge or discharge

- Figure 4.1a shows the schematic for a simple RC circuit. Simulate the circuit using PSpice as per the case I in Table 4.1.
- Specify the "Run to Time" as 100 ms to run the simulation for five full periods.

Pre Lab 04

Q.2) Natural Response: of a circuit is how it behaves with no external force (input). It only depends on energy initially stored.

RL circuit: $i(t) = I_0 e^{-t/\tau}$ for $\tau = L/R$

RC circuit: $v(t) = V_0 e^{-t/\tau}$ for $\tau = RC$

Forced Response: of a circuit is how it behaves when driven by an external input also known as steady-state response. It shows circuit's behavior to the input in long run.

RL circuit: $i_t = V_{in} / R$

RC circuit: $V_t = V_{in}$

Complete Response: is the sum of natural response and forced response. It describes how circuit behaves from initial time until it reaches steady state.

RL Circuit: $i(t) = i_f + (i_0 - i_f) e^{-t/\tau}$ for $\tau = L/R$

RC Circuit: $v(t) = V_f + (V_0 - V_f) e^{-t/\tau}$ for $\tau = RC$